Simplify algebraic fractions

1 Simplify:

a $\dfrac{40y}{450}$ **b** $\dfrac{6n}{9n}$ **c** $\dfrac{42xy}{24y}$ **d** $\dfrac{15n^2}{-40n}$

e $\dfrac{64mn^2}{40m^2n}$ **f** $-\dfrac{10x^2}{6xy^3}$ **g** $\dfrac{6p^5}{3p^3}$ **h** $\dfrac{10a^3b^5}{25a^2b^9}$

2 Simplify:

a $\dfrac{25x-40}{55}$ **b** $\dfrac{4uv-20uw}{4u}$ **c** $\dfrac{5u}{5uv-30uw}$ **d** $\dfrac{9m^2+24mn}{18m^2}$

3 Simplify:

a $\dfrac{4x+20y}{x+5y}$ **b** $\dfrac{2x-6y}{x-3y}$ **c** $\dfrac{x-2}{2-x}$ **d** $\dfrac{5y-30}{6-y}$

e $\dfrac{2x+6}{x^2+3x}$ **f** $\dfrac{8-2x}{5x^2-20x}$ **g** $\dfrac{10x-15}{2x^3-3x^2}$ **h** $\dfrac{8-2xy}{x^2y-4x}$

i $\dfrac{12m^2+6mn}{18mn+36m^2}$

4 Simplify:

a $\dfrac{(x+3)(x-2)}{x+3}$ **b** $\dfrac{(2x-5)(x+4)}{x+4}$ **c** $\dfrac{x-7}{2(x-7)(x+4)}$ **d** $\dfrac{(x+6)(x+8)}{(x+6)(x-9)}$

e $\dfrac{x^2-8x+16}{x-4}$ **f** $\dfrac{x^2+7x+12}{x+4}$ **g** $\dfrac{x^2+9x+20}{(x+5)(x+1)}$ **h** $\dfrac{x^2-3x-18}{(x-6)(x+2)}$

5 Simplify:

a $\dfrac{m^2-16}{m-4}$ **b** $\dfrac{m^2-64}{8+m}$ **c** $\dfrac{x-10}{x^2-100}$ **d** $\dfrac{a^2-121}{11-a}$

6 Simplify:

a $\dfrac{8(p-q)^2}{8p^2-8q^2}$ **b** $\dfrac{x+10}{x^2+20x+100}$ **c** $\dfrac{x^2-4x+4}{7(x-2)}$ **d** $\dfrac{x^2+8x+15}{x+5}$

e $\dfrac{x+5}{x^2+2x-15}$ **f** $\dfrac{3x^2-18x-216}{4(x+6)}$ **g** $\dfrac{3y+6}{y^2-4}$ **h** $\dfrac{m^2-16}{5m+20}$

i $\dfrac{x^2+6x+9}{x^2-9}$ **j** $\dfrac{x^2-3x-10}{x^2-5x}$ **k** $\dfrac{x^2+5x+6}{x^2-2x-8}$ **l** $\dfrac{8(x+3)}{5x^2+30x+45}$

m $\dfrac{(x+3)(x-11)}{x^2-18x+77}$ **n** $\dfrac{4x^2+8x-192}{2x+16}$

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7 Simplify:

a $\dfrac{xy+10x+9y+90}{x+9}$

b $\dfrac{5(x-12)}{xy+6x-12y-72}$

c $\dfrac{y-8}{3xy-24x-30y+240}$

d $\dfrac{5xy-20x+40y-160}{8(y-3)(y-4)}$

e $\dfrac{wxy+wxz}{wy+wz+xy+xz}$

f $\dfrac{x^2-xy+xz-yz}{x^2-2xy+y^2}$

8 Simplify the following algebraic fraction:

$$\frac{5\left(k^2-5\right)^4+20k\left(k^2-5\right)^5}{15\left(k^2-5\right)^4}$$

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